Project: Worked examples
Title: Plane wall heat transfer (layers and films)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Steady heat flow through a composite plane wall between two fluids, or between two known surface temperatures: each layer's conduction resistance, each film's convection resistance, the overall U-value, the heat flow and flux, and every surface and interface temperature.

WHAT IT CALCULATES	RELATION
Resistance of each surface film	$R = 1/(h \cdot A)$
Resistance of each conducting layer	$R_i = t_i/(k_i \cdot A)$
Total resistance, hot side first	$R_{tot} = \Sigma R$
Heat flow through the wall	$Q = (T_1 - T_2)/R_{tot}$
Heat flux and overall U-value	$q = Q/A$; $U = 1/(R_{tot} \cdot A)$
Surface and interface temperatures, stepping down from T_1 across each resistance in turn	$T_{next} = T_{previous} - Q \cdot R$

Basis: The plane wall: thermal resistance and the composite wall[1](§3.1); Generalized thermal resistance networks (multilayer plane walls)[2](§3-3); Thermal resistance of building components (method)[3](§6).

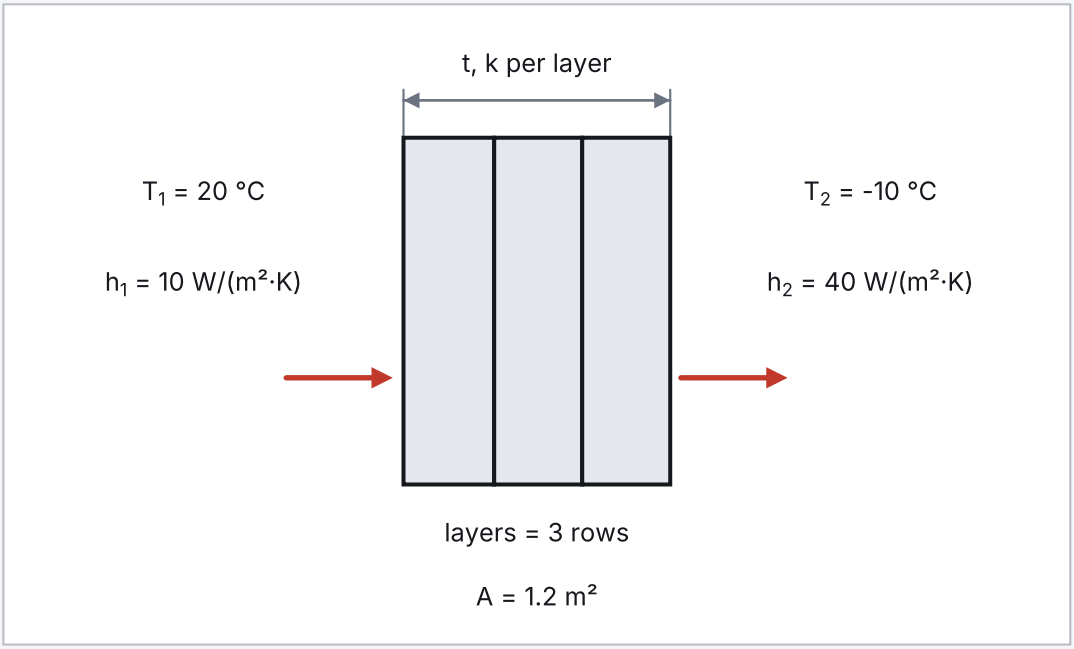


Figure 1 — A composite plane wall, side 1 left: a fluid film, the layers as listed (three drawn), a film on side 2. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
T_1	Side 1 temperature (the fluid's, or the surface's when no film is given)	20	°C	spec: SPEC-BS-04 design conditions	-60 … 400 °C

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
h_1	Side 1 film coefficient (omit when T_1 is the surface temperature)	10	W/(m²·K)	spec: SPEC-BS-04 design conditions	2 … 10000 W/(m²·K)
layers	Conducting layers, side 1 first	3 row(s)		drawing: DRG-GL-802 rev A	any
h_2	Side 2 film coefficient (omit when T_2 is the surface temperature)	40	W/(m²·K)	spec: SPEC-BS-04 design conditions	2 … 10000 W/(m²·K)
T_2	Side 2 temperature (the fluid's, or the surface's when no film is given)	-10	°C	spec: SPEC-BS-04 design conditions	-60 … 400 °C
A	Wall area normal to the heat flow	1.2	m²	drawing: DRG-GL-802 rev A	0.001 … 10000 m²

TABLE ROW	VALUES
layers[1]	name = glass; t = 4 mm; k = 0.78 W/(m*K)
layers[2]	name = air gap; t = 10 mm; k = 0.026 W/(m*K)
layers[3]	name = glass; t = 4 mm; k = 0.78 W/(m*K)

3. Assumptions

A1. Steady, one-dimensional conduction through plane layers of constant conductivity, with no internal heat generation. (Thermal-circuit idealisation: equal layer areas, no edge losses.)
A2. Layers are in perfect thermal contact; each film coefficient is constant over its surface. (A contact resistance can be entered as a thin layer of equivalent thickness and conductivity.)

4. Working

[6] $R_{h1} = 1 / (h_1 \cdot A)$ [1](§3.1)
$= 1 / (10\ \text{W/(m²·K)} \times 1.2\ \text{m²})$
$= 0.083333\ \text{K/W}$
Side 1 film resistance

[7] $R_1 = t_1 / (k_1 \cdot A)$ [1](§3.1)
$= 4\ \text{mm} / (0.78\ \text{W/(m·K)} \times 1.2\ \text{m²})$
$= 0.0042735\ \text{K/W}$
Conduction resistance of glass

[8] $R_2 = t_2 / (k_2 \cdot A)$ [1](§3.1)
$= 10\ \text{mm} / (0.026\ \text{W/(m·K)} \times 1.2\ \text{m²})$
$= 0.32051\ \text{K/W}$
Conduction resistance of air gap

[9] $R_3 = t_3 / (k_3 \cdot A)$ [1](§3.1)
$= 4\ \text{mm} / (0.78\ \text{W/(m·K)} \times 1.2\ \text{m²})$
$= 0.0042735\ \text{K/W}$
Conduction resistance of glass

[10] $R_{h2} = 1 / (h_2 \cdot A)$ [1](§3.1)
$= 1 / (40\ \text{W/(m²·K)} \times 1.2\ \text{m²})$
$= 0.020833\ \text{K/W}$
Side 2 film resistance

[11] $R_{tot} = \Sigma R$ [2](§3-3)
$= 0.083333 + 0.0042735 + 0.32051 + 0.0042735 + 0.020833\ \text{K/W}$
$= 0.43323\ \text{K/W}$

Total thermal resistance

$$[12]\ Q = (T_1 - T_2) / R_{tot}$$
$$= (20\ degC - (-10\ degC)) / 0.43323\ K/W$$
$$= 69.248\ W$$

Heat flow from side 1 to side 2

$$[13]\ q = Q / A$$
$$= 69.248\ W / 1.2\ m^2$$
$$= 57.707\ W/m^2$$

Heat flux

$$[14]\ U = 1 / (R_{tot} * A)$$
$$= 1 / (0.43323\ K/W \times 1.2\ m^2)$$
$$= 1.9236\ W/(m^2 \cdot K)$$

Overall heat transfer coefficient

$$[15]\ T_{s0} = T_1 - Q \cdot R_{h1}$$
$$= 20\ °C - 69.248\ W \times 0.083333\ K/W$$
$$= 14.229\ degC$$

Side 1 surface

$$[16]\ T_{s1} = T_{s0} - Q \cdot R_1$$
$$= 14.229\ °C - 69.248\ W \times 0.0042735\ K/W$$
$$= 13.933\ degC$$

Interface 1

$$[17]\ T_{s2} = T_{s1} - Q \cdot R_2$$
$$= 13.933\ °C - 69.248\ W \times 0.32051\ K/W$$
$$= -8.2614\ degC$$

Interface 2

$$[18]\ T_{s3} = T_{s2} - Q \cdot R_3$$
$$= -8.2614\ °C - 69.248\ W \times 0.0042735\ K/W$$
$$= -8.5573\ degC$$

Side 2 surface

5. Results

SYMBOL	VALUE	DESCRIPTION
Q	69.2 W	Heat flow
q	57.7 W/m²	Heat flux
U	1.92 W/(m²·K)	Overall heat transfer coefficient
R_{tot}	0.433 K/W	Total thermal resistance
R_{h1}	0.083 K/W	Side 1 film resistance
R_1	0.0043 K/W	Layer 1 resistance
R_2	0.321 K/W	Layer 2 resistance
R_3	0.0043 K/W	Layer 3 resistance
R_{h2}	0.021 K/W	Side 2 film resistance
T_{s0}	14.2 °C	Side 1 surface temperature
T_{s1}	13.9 °C	Interface 1 temperature
T_{s2}	-8.3 °C	Interface 2 temperature
T_{s3}	-8.6 °C	Side 2 surface temperature

6. Checks (governing first)

None.

7. Applicability

None.

8. Verification

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Template bs.plane-wall@1.0.1 Content hash d3270684be534c78 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. Bergman TL, Lavine AS, Incropera FP, DeWitt DP. Fundamentals of Heat and Mass Transfer. 7th ed. Wiley; 2011.
2. Çengel YA, Ghajar AJ. Heat and Mass Transfer: Fundamentals and Applications. 5th ed. McGraw-Hill Education; 2015.
3. British Standards Institution. Building Components and Building Elements—Thermal Resistance and Thermal Transmittance—Calculation Methods. BS EN ISO 6946. BSI.